

UI/UX & Frontend Features

Master Reference Guide

A practical glossary of 23 interface categories, plus a learning map for UI/UX, frontend development, and AI-assisted coding.

Will one UI/UX course teach all 23 categories?

Not completely. A broad UI/UX course can teach the names, purposes, visual design, user flows, accessibility concepts, and common patterns. It usually will not teach you how to code every feature, connect databases and APIs, upload files securely, implement authentication, or deploy a working app. For that, you also need frontend and basic full-stack development.

What each subject teaches

- **UI design:** Visual hierarchy, spacing, typography, color, layouts, components, states, responsive design, and design systems.
- **UX design:** User research, information architecture, user flows, usability, interaction patterns, accessibility, testing, and product decisions.
- **Frontend development:** HTML, CSS, JavaScript, React, components, state, forms, browser behavior, accessibility implementation, APIs, and responsive interfaces.
- **Backend / full-stack development:** Databases, authentication, file storage, permissions, server logic, security, payments, and deployment.
- **Vibe coding / AI-assisted development:** Using Codex or Claude Code to plan, generate, inspect, debug, test, and improve the implementation. It accelerates the work but does not replace understanding the categories above.

Recommended YouTube learning sequence

1. UI UX terminology every designer should know
2. common UI components explained
3. UI design patterns full course
4. design systems full course
5. Figma UI UX full course
6. visual hierarchy spacing typography UI design
7. frontend development full course HTML CSS JavaScript React
8. accessible UI components ARIA full course
9. vibe coding full course Claude Code
10. Claude Code beautiful frontend Next.js shadcn
11. OpenAI Codex frontend development full course
12. how web apps work frontend backend database API authentication

1. Navigation

Feature / term	What it does
Navbar	Main navigation across the top
Sidebar	Vertical navigation panel
Bottom navigation	Mobile navigation at the bottom
Breadcrumbs	Shows location in a hierarchy
Tabs	Switches between related views
Hamburger menu	Collapses navigation on small screens
Mega menu	Large grouped navigation menu
Dropdown menu	Reveals more links or actions
Context menu	Actions relevant to a selected item
Command palette	Searchable launcher for actions
Pagination	Moves between result pages
Infinite scroll	Loads more content while scrolling
Anchor links	Jumps to a page section
Table of contents	Links to document sections
Stepper	Shows progress through a sequence
Wizard	Guided multi-step process

2. Search and discovery

Feature / term	What it does
Search bar	Accepts a search query
Autocomplete / typeahead	Suggests results while typing
Combobox	Searchable input with selectable options
Fuzzy search	Handles partial matches and misspellings
Faceted search	Filters across multiple categories
Filter chips	Shows active filters as removable labels
Sort control	Changes result order
Advanced search	Builds detailed search conditions
Recent searches	Shows previous queries
Saved search	Stores a query for reuse
Search highlighting	Emphasizes matching text
Global search	Searches across the whole application
Entity picker	Searches and selects people, schools, files, etc.
Location search	Suggests places or addresses
Date-range picker	Filters using start and end dates

3. Inputs and forms

Feature / term	What it does
Text input	Accepts short text
Textarea	Accepts long text
Select	Chooses one option
Multi-select	Chooses several options
Checkbox	Toggles independent choices

Radio group	Selects one option from a group
Toggle / switch	Turns a setting on or off
Slider	Selects a numeric range
Number input	Accepts numeric values
Password input	Masks secret text
Date picker	Selects a date
Time picker	Selects a time
File picker	Chooses a local file
Dropzone	Accepts dragged-and-dropped files
Rich-text editor	Formats text visually
Markdown editor	Edits Markdown
Code editor	Edits source code
Tag input	Adds several short values
OTP input	Enters a verification code
Signature pad	Captures a handwritten signature
Rating input	Records a score
CAPTCHA	Checks whether the user is human
Form wizard	Breaks a long form into steps

4. Buttons and actions

Feature / term	What it does
Primary button	Main call to action
Secondary button	Supporting action
Tertiary button	Low-emphasis action
Icon button	Action represented by an icon
Floating action button	Prominent floating action
Split button	Default action plus alternatives
Button group	Groups related actions
Segmented control	Switches between related modes
Destructive button	Deletes or removes content
Copy button	Copies to the clipboard
Share button	Opens sharing options
Download / export	Downloads data or files
Upload / import	Adds files or data
Undo / redo	Reverses or restores actions
Bulk actions	Applies one action to many items
Overflow menu	Stores less common actions under a more menu

5. Feedback and status

Feature / term	What it does
Toast / snackbar	Temporary notification
Alert / banner	Important persistent message
Inline message	Feedback near the relevant control
Success state	Confirms completion
Error state	Explains a failure

Warning state	Signals possible risk
Loading spinner	Shows ongoing work
Progress bar	Shows completion percentage
Skeleton loader	Represents content while loading
Empty state	Explains what to do when no data exists
No-results state	Appears when search finds nothing
Offline state	Shows lost connectivity
Retry state	Allows a failed action to run again
Autosave indicator	Shows whether changes are saved
Sync indicator	Shows synchronization status
Badge	Displays a compact status or count
Status dot	Shows presence or availability

6. Overlays and temporary surfaces

Feature / term	What it does
Modal / dialog	Focused task above the page
Confirmation dialog	Requests confirmation
Drawer / side sheet	Slides in from the side
Bottom sheet	Slides up from the bottom
Popover	Contextual floating panel
Tooltip	Brief explanatory hint
Lightbox	Enlarged media viewer
Full-screen overlay	Temporarily fills the viewport
Cookie banner	Requests consent
Notification center	Collects alerts
Product tour	Guides users through features
Spotlight	Highlights one interface element

7. Content containers

Feature / term	What it does
Card	Groups related information
Panel	Contains a larger section
Accordion	Expands and collapses content
Collapsible	Hides optional content
Carousel	Rotates through items
Gallery	Displays media in a grid
List	Shows repeated vertical items
Grid	Arranges items in rows and columns
Table	Displays structured data
Tree view	Shows nested items
Timeline	Displays chronological events
Feed	Shows continually updated content
Kanban board	Organizes cards into columns
Dashboard	Summarizes key information
Hero section	Highlights the main message

Callout	Emphasizes important information
Divider	Separates content
Avatar	Represents a user
Chip / pill	Compact label or selection
Tag	Categorizes content

8. Tables and data-heavy interfaces

Feature / term	What it does
Data table	Displays structured records
Sortable column	Reorders table data
Filterable column	Filters one field
Resizable column	Adjusts column width
Pinned column	Stays visible while scrolling
Sticky header	Keeps headings visible
Row selection	Selects records
Expandable row	Reveals more detail
Inline editing	Edits data directly
Bulk selection	Selects multiple rows
Column chooser	Shows or hides columns
Virtualized list	Efficiently renders large datasets
Grouped rows	Organizes records by category
Summary row	Shows totals or aggregates
Export control	Exports table data
Density control	Changes row spacing

9. File and document features

Feature / term	What it does
File uploader	Uploads files
Dropzone	Accepts dragged files
Upload queue	Lists pending files
Upload progress	Shows transfer progress
File preview	Displays a file before submission
PDF viewer	Displays PDFs inside the app
Image cropper	Adjusts image framing
File browser	Navigates stored files
Attachment list	Shows attached documents
URL importer	Imports content from a link
Document scanner	Captures documents with a camera
File-type validation	Rejects unsupported formats
File-size validation	Rejects oversized files
Replace file	Swaps an uploaded file
Version history	Shows earlier versions
Annotation tools	Adds highlights or comments

10. Authentication and accounts

Feature / term	What it does
Sign-in form	Authenticates a user
Sign-up form	Creates an account
Social login	Signs in with Google, Apple, etc.
Password reset	Restores account access
Email verification	Confirms email ownership
Two-factor authentication	Adds a second security factor
Passkey login	Uses device-based authentication
Session timeout	Ends inactive sessions
Profile page	Shows personal details
Account settings	Manages preferences
Role selector	Assigns access levels
Device management	Manages active sessions

11. E-commerce features

Feature / term	What it does
Product card	Displays an item
Product gallery	Shows product images
Variant selector	Chooses size, color, etc.
Add to cart	Adds an item to the cart
Shopping cart	Stores selected products
Mini-cart	Compact cart preview
Checkout flow	Completes a purchase
Quantity selector	Changes item quantity
Wishlist	Saves products
Comparison table	Compares products
Promo-code input	Applies a discount
Order tracker	Shows shipment progress
Stock indicator	Shows availability
Price-range slider	Filters by price
Reviews and ratings	Displays customer feedback
Subscription selector	Chooses billing frequency

12. Social and communication features

Feature / term	What it does
Comment box	Adds a comment
Threaded comments	Organizes nested replies
Like button	Records approval
Reaction picker	Chooses a reaction
Share sheet	Shares content
Follow button	Subscribes to an account
Mention input	Tags another user
Hashtag input	Adds a topic label

Direct messages	Sends private messages
Chat window	Supports live conversation
Typing indicator	Shows that someone is typing
Read receipt	Shows message status
Presence indicator	Shows online availability
Notification bell	Opens notifications
Activity feed	Shows recent actions
Composer	Creates posts
Poll	Collects votes

13. Scheduling and calendar features

Feature / term	What it does
Calendar view	Displays events
Agenda view	Lists upcoming events
Date-range picker	Chooses a date interval
Time-slot picker	Selects an available appointment
Recurrence control	Creates repeating events
Availability grid	Shows free and busy periods
Event card	Displays event details
Drag-to-reschedule	Moves an event
Time-zone selector	Changes time zone
Reminder control	Sets notifications
Booking flow	Schedules an appointment
Countdown	Shows time remaining

14. Maps and location interfaces

Feature / term	What it does
Interactive map	Displays geographic data
Map marker	Marks a location
Marker cluster	Groups nearby markers
Geolocation button	Finds the user's position
Address autocomplete	Suggests addresses
Route planner	Calculates directions
Heatmap	Visualizes density
Geofence editor	Defines a geographic area
Map layers	Toggles map information
Street view	Shows ground-level imagery
Distance calculator	Measures distance
Location card	Summarizes a place

15. Media features

Feature / term	What it does
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Video player	Plays video
Audio player	Plays audio
Scrubber	Moves through media
Volume control	Adjusts sound
Captions	Displays spoken dialogue
Playback-speed control	Changes speed
Picture-in-picture	Uses a floating player
Full-screen control	Expands media
Playlist	Organizes media
Thumbnail strip	Navigates media items
Waveform	Visualizes audio
Recording control	Records audio or video
Livestream badge	Shows live status
Transcript panel	Displays searchable dialogue

16. Editing and creation tools

Feature / term	What it does
Rich-text editor	Formats text
Markdown editor	Creates Markdown
Code editor	Edits code
Drag-and-drop builder	Builds layouts visually
Canvas	Positions visual objects
Toolbar	Groups editing actions
Formatting menu	Changes text styles
Block editor	Builds content from movable blocks
Slash command	Inserts content using /
Inline toolbar	Appears near selected content
Undo history	Reverses edits
Autosave	Saves continuously
Version history	Restores earlier revisions
Live preview	Shows output while editing
Collaboration cursors	Shows other editors
Comments	Adds review feedback
Track changes	Records suggested edits

17. Onboarding and education

Feature / term	What it does
Welcome screen	Introduces the product
Onboarding checklist	Guides setup
Product tour	Explains key features
Coach marks	Points to controls
Walkthrough	Demonstrates a process
Sample data	Shows how the product works
Empty-state guidance	Suggests the first action
Progress tracker	Shows setup completion

Contextual help	Explains features in place
Help center	Contains support articles
Interactive tutorial	Teaches through actions
Feature announcement	Introduces updates

18. Personalization and settings

Feature / term	What it does
Settings page	Configures the application
Theme selector	Switches appearance
Language selector	Changes locale
Notification preferences	Controls alerts
Privacy controls	Manages data settings
Dashboard customization	Rearranges widgets
Saved views	Stores filters and layouts
Favorites	Saves frequently used items
Recent items	Shows activity history
Recommendations	Suggests relevant content
Accessibility settings	Adjusts contrast, text size, motion, etc.

19. Accessibility features

Feature / term	What it does
Keyboard navigation	Supports use without a mouse
Focus indicator	Shows the active element
Skip link	Bypasses repeated navigation
Accessible name	Labels controls for assistive technology
ARIA live region	Announces dynamic updates
High-contrast mode	Improves visibility
Text resizing	Enlarges content
Reduced motion	Minimizes animation
Captions	Makes audio readable
Transcript	Provides a text alternative
Error summary	Lists form errors
Accessible combobox	Supports search and selection accessibly
Focus trap	Keeps keyboard focus inside a dialog

20. Mobile-specific patterns

Feature / term	What it does
Bottom navigation	Primary mobile navigation
Bottom sheet	Mobile contextual panel
Pull to refresh	Reloads by swiping
Swipe actions	Reveals actions with gestures
Floating action button	Prominent mobile action

Long-press menu	Opens touch context actions
Mobile drawer	Displays navigation
Sticky action bar	Keeps actions visible
Thumb-friendly controls	Places controls within reach
Haptic feedback	Confirms touch actions
Camera upload	Captures images directly
Biometric login	Uses face or fingerprint authentication

21. AI-specific interface features

Feature / term	What it does
Prompt box / composer	Accepts AI instructions
Chat interface	Supports conversational interaction
Streaming response	Displays output while generated
Stop button	Interrupts generation
Regenerate button	Produces another answer
Suggested prompts	Offers prompt starters
Prompt templates	Stores reusable instructions
Model selector	Changes the AI model
Tool selector	Enables capabilities
File attachment	Adds contextual files
Citation display	Shows supporting sources
Processing indicator	Shows ongoing reasoning or execution
Editable output	Allows direct revision
Version comparison	Compares generated outputs
Feedback controls	Rates responses
Usage meter	Shows context or credit usage
Agent activity log	Displays completed actions
Approval step	Requires confirmation before an action
Retrieval panel	Shows retrieved sources
Structured output	Returns cards, tables, or fields
Voice input	Accepts spoken prompts

22. Interaction concepts

Feature / term	What it does
Hover state	Appearance under the pointer
Focus state	Appearance during keyboard focus
Active state	Appearance while pressed
Disabled state	Unavailable control
Selected state	Current choice
Loading state	State while fetching data
Optimistic update	Updates before server confirmation
Debouncing	Waits until input pauses
Throttling	Limits action frequency
Lazy loading	Loads only when needed
Progressive disclosure	Reveals complexity gradually

Inline validation	Validates near the input
Responsive design	Adapts to screen size
Adaptive design	Uses device-specific layouts
Microinteraction	Small response to an action
Drag and drop	Moves or uploads by dragging
Direct manipulation	Interacts with visible objects
Autosave	Saves without a manual action
Undoable action	Allows reversal
Keyboard shortcut	Performs actions with keys
Deep linking	Links to a specific app state
State persistence	Retains progress or settings
Role-based UI	Changes by user permissions

23. Design-system vocabulary

Feature / term	What it does
Design system	Shared rules, components, and styles
Component library	Reusable interface components
Pattern library	Common interaction solutions
Design tokens	Stored values for color, spacing, typography, etc.
Variant	Alternative component version
Component state	Loading, disabled, selected, error, etc.
Primitive	Low-level reusable component
Headless component	Logic and accessibility without fixed styling
Composition	Combining small components
Responsive breakpoint	Width where the layout changes
Grid system	Rules for alignment
Spacing scale	Standard spacing increments
Typography scale	Standard text sizes
Icon set	Consistent icon collection
Visual hierarchy	Relative emphasis of elements
Information architecture	Organization of content and navigation

The 30 terms to learn first

Autocomplete, Combobox, Entity picker, Fuzzy search, Debounced search, **Dropzone**, File uploader, File preview, URL importer, Modal, **Drawer**, Popover, Toast, Tabs, Accordion, **Multi-step form**, Inline validation, Empty state, Skeleton loader, Data table, **Filter chips**, Command palette, Dashboard, Rich-text editor, Drag-and-drop sorting, **Responsive design**, Accessibility, Design system, Component state, Visual hierarchy.

Prompt examples for Codex or Claude Code

College lookup

Build an accessible college entity picker using a combobox with 250 ms debounced fuzzy search. Show matching colleges in a keyboard-navigable dropdown. Support click, arrow keys, Enter, and Escape. Display selected colleges as removable chips. Include loading, no-results, error, and disabled states.

PDF or link import

Build a document import card with a drag-and-drop PDF dropzone, browse button, and separate URL field. Validate file type and size, show upload progress, preview the selected PDF, and allow removal or replacement. Include accessible labels, keyboard operation, and clear error messages.

Professional UI review

Audit this page for visual hierarchy, spacing, typography, responsiveness, interaction states, accessibility, empty/loading/error states, and consistency with the design system. List the problems first, then implement fixes without changing the product requirements.